Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/13/2004
PI Name:	Goddard, Joe D. Ph.D.		
Project Title:	Vibratory Dynamics and Transport of Granular Media at Various G Levels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
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Zip Code:	92093-0411	**Congressional District:**	49
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2465		
Performance Goal No.:			
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Task Description:	This is a four-year program of research on the vibratory dynamics of granular media. The research will make use of kinetic-theory based continuum hydrodynamic models to predict the behavior of granular beds subject to vibratory excitation at various g levels. This task involves the development of continuum models which are capable of capturing both the fluid-like behavior of granular layers in free flight and their solid-like behavior during impulsive contact and sliding on solid surfaces. The intent is: 1. To develop a sound continuum-mechanical basis for interpreting recent scientific studies on vertically vibrated granular layers. 2. To extend the calculations to more complex and technologically important granular flows in vibratory beds and		

conveyors.

Part of the effort will be aimed at the design of ground-based laboratory experiments on a model vibratory apparatus to test the theory. Similar experiments will be designed for a flight-based NASA Space Shuttle Glovebox experiment. This effort, which will begin in the third year of the grant, will depend on other sources of funding for its realization.

The overall objective of the research is to provide an enhanced understanding of the mechanics of granular flow and to apply the knowledge to rationalize and extend various engineering methodologies for vibratory conveying of granular solids in various g-level environments.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Enhanced understanding of granular mechanics and vibratory conveying.

Task Progress:

The principal research associate, Dr. Anjani Didwania, has maintained a somewhat independent but related research activity and collaboration on the mechanics of fluid particle systems, ranging from instabilities in fluidized beds to the theory of liquid-saturated porous media.
Partial support from the current NASA grant is cited in essentially all the listed publications and talks.

The following narrative provides a brief summary of progress in three principal areas of research activity, 1) The mechanics of vibrated granular layers and vibratory conveying, 2) Material and dynamical instabilities of granular flows and fluid-particle systems, 3) Modeling of discrete systems and heterogeneous continua, and 4) Interactions and collaboration with other research groups in the mechanics of granular media and heterogeneous continua.

1. The mechanics of vibrated granular layers and vibratory conveying

As the first and principal effort on of he task, a continuum model has been developed to describe various wavy ("Faraday") patterns on shallow vibrated granular layers driven by sinusoidal vertical motion of a rigid flat bottom. Above a certain plate acceleration threshold, this motion is sustained by a succession of normal impacts between plate and granular layer.

In the proposed model, the layer is treated as a viscous liquid in free flight, punctuated by impulsive solid-like collisional "restitution" and rebound at the bottom boundary. Essential to the model is an impulsive line-force singularity, the "stosslet," leading to a set of nonlinear partial differential equation (PDE) with line impulses originating from bottom collision points.

We have developed numerical solutions for nearly inviscid layers, with a small viscosity serving to regularize otherwise discontinuous states. We have investigated the evolution from initially perturbed flat layers into various patterned states. In particular, we find “kink” and “oscillon” structure that bear a striking resemblance to those observed experimentally. We have also performed a stability analysis on bouncing flat layers and have been able to show that, unlike liquid layers, “Faraday” patterns cannot be due to a linear free-surface instability of flat layers, contrary to the published findings of previous investigators. Instead, the instability appears to be of the non-linear variety, arising from the impact with the vibrating plate. This work is the subject of a paper which has been submitted for publication and is still under review.

It appears possible to extend above model to vibratory conveying, where the granular layer undergoes a complex cycle of oblique impact with frictional sliding on a vibrating conveyor surface. Hence, as a continuing effort, we are developing a description of oblique impact with frictional sliding, which is characteristic of the "throw" mode of vibratory conveying. In this connection, some initial consideration has been given to the type of closed conduit thought to be appropriate to microgravity conveying, and an exact solution has been found for one basic model of slide conveying. This latter is the subject of a paper in preparation, which is virtually completed. An extensive review paper on vibratory conveying was begun but has been put on hold in favor of the above research activities dealing with fundamental mechanics issues.

The subject of shear flow instability, mentioned next, is relevant to vibratory conveying, since shearing impacts are involved.

2. Material and dynamical instabilities of granular flows and fluid-particle systems

As a completion and continuation of work done on the prior NASA grant a number of papers have been started or completed on the subject of material and flow instabilities in granular and particulate systems. The review article on material instability (Goddard JD."Material Instability in Complex Fluids", Ann. Rev. Fluid Mech., 35, 113-33, 2003) provides a synthesis of a large body of literature on material instability, and serves to distinguish it from the usual dynamical instabilities, of a type that are probably implicated in the mechanics of vibrated layers.

3. Modeling of discrete systems and heterogeneous continua

The balance of the work cited below deals with continuum modeling issues for granular materials and other heterogeneous continua, including fluid-particle systems and fluid-saturated media. Many of these issues are common to micromechanical and kinetic theories of granular materials.

4. Research Interactions

Part of the cited work represents an ongoing collaboration with Drs. Prabhu Nott and M. Alam, of the Indian Institute of Science in Bangalore, India. Both received financial support from the previous NASA grant, Dr. Nott as a visiting professor, and Dr. Alam, as a postdoctoral researcher. Dr. Alam continued his work on granular mechanics during subsequent postdoctoral appointment as Humboldt Fellow at the Institute of Computer Applications, in the University of Stuttgart, before accepting a faculty position at Bangalore.

The work on fluid-saturated porous media represents a long-standing collaboration between Dr. Didwania and Professor R. DeBoer, of the Dept. of Civil Engineering, in the University of Essen, Germany. This collaboration began under the prior NASA grant, with two extended visits of Prof. DeBoer to our research group, supported by his institution. Support for related international travel has been supported by a National Science Foundation travel grant to Dr. Didwania.

The principal investigator (PI) spent periods 8-29 September and 1 November-30 December 2002 at the University of Paris VI in the research group of Dr. Eric Clement, a widely recognized French researcher in granular mechanics, who has performed experiment and numerical simulations of vibrated granular layers. He is also spent the period 30 September to 25 October 2002 as a guest of the Max Planck Institute for the Physics of Complex Systems, Dresden Germany, as a participant in the program “Geometry and Mechanics of Structured Materials (GEOMES).” There he worked with various researchers on the mechanics of particulate cellular structures, including an on going collaboration with one participant on the stress-controlled remolding of human bone. As part of the European stay, the PI presented seminars at the Universities of Paris and Montpellier where he also served on two doctoral dissertation committees on granular mechanics.

Summary

The work carried out in the course of this investigation has set the stage for further modeling efforts on the topic of main concern, vibratory mechanics and conveying of granular materials. Among the several important needs identified but not satisfied by the current effort are:

1. Simple kinetic theories of dense granular systems, including the effect of granular temperature induced by impact with solid surfaces on phenomenological parameters such as the effective coefficient of restitution and viscosity of a granular mass.

2. More powerful numerical techniques for treating the continuum mechanics of granular masses undergoing free flight punctuated by impact with solid surfaces.

3. Solutions to the equations developed in this work for vibrated granular layers that would describe the spatially periodic patterns that predominate in experiments.

4. Crucial experiments on vibrated layers in vacuo, to measure properties observed only qualitatively in previous experiments by others (e.g., oscillon and kink dimensions).

As reflected in prior progress reports, Item 2 turned out to be a major and time-consuming activity in the present work, whereas the level of support afforded by the current NASA grant did not allow for consideration of Item 4. Some of these issues and needs will be addressed in continuing research, some of which is the subject of a future research proposal to NASA.

Bibliography Type:	Description: (Last Updated: 07/23/2019)
Articles in Peer-reviewed Journals	Goddard JD. "On entropy estimates of contact forces in static granular assemblies." International Journal of Solids and Structures. 2004 Oct;41(2);5851-61. https://doi.org/10.1016/j.ijsolstr.2004.05.049 , Oct-2004
Articles in Peer-reviewed Journals	de Boer R, Didwania AK. "Two phase flow and the capillarity phenomenon in porous solids - a continuum thermomechanical approach." Transport in Porous Media. 2004 Aug;56(2):137-70. https://doi.org/10.1023/B:TIPM.0000021731.14083.7f , Aug-2004
Articles in Peer-reviewed Journals	Alam M, Arakeri VH, Nott PR, Goddard JD. "Instability-induced ordering, universal unfolding and the role of gravity in granular Couette flow." Journal of Fluid Mechanics. 2005 Jan;523:277-306. https://doi.org/10.1017/S0022112004002150 , Jan-2005